2019 HIGHER SCHOOL CERTIFICATE EXAMINATION

# Physics

## DATA SHEET

| | |
|---|---|
| Charge on electron, $q_e$ | $-1.602 \times 10^{-19}$ C |
| Mass of electron, $m_e$ | $9.109 \times 10^{-31}$ kg |
| Mass of neutron, $m_n$ | $1.675 \times 10^{-27}$ kg |
| Mass of proton, $m_p$ | $1.673 \times 10^{-27}$ kg |
| Speed of sound in air | $340\ \mathrm{m\,s^{-1}}$ |
| Earth's gravitational acceleration, $g$ | $9.8\ \mathrm{m\,s^{-2}}$ |
| Speed of light, $c$ | $3.00 \times 10^{8}\ \mathrm{m\,s^{-1}}$ |
| Electric permittivity constant, $\varepsilon_0$ | $8.854 \times 10^{-12}\ \mathrm{A^2\,s^4\,kg^{-1}\,m^{-3}}$ |
| Magnetic permeability constant, $\mu_0$ | $4\pi \times 10^{-7}\ \mathrm{N\,A^{-2}}$ |
| Universal gravitational constant, $G$ | $6.67 \times 10^{-11}\ \mathrm{N\,m^2\,kg^{-2}}$ |
| Mass of Earth, $M_E$ | $6.0 \times 10^{24}$ kg |
| Radius of Earth, $r_E$ | $6.371 \times 10^{6}$ m |
| Planck constant, $h$ | $6.626 \times 10^{-34}$ J s |
| Rydberg constant, $R$ (hydrogen) | $1.097 \times 10^{7}\ \mathrm{m^{-1}}$ |
| Atomic mass unit, $u$ | $1.661 \times 10^{-27}$ kg<br>$931.5\ \mathrm{MeV}/c^2$ |
| 1 eV | $1.602 \times 10^{-19}$ J |
| Density of water, $\rho$ | $1.00 \times 10^{3}\ \mathrm{kg\,m^{-3}}$ |
| Specific heat capacity of water | $4.18 \times 10^{3}\ \mathrm{J\,kg^{-1}\,K^{-1}}$ |
| Wien's displacement constant, $b$ | $2.898 \times 10^{-3}$ m K |



## FORMULAE SHEET

### Motion, forces and gravity

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

$$\Delta U = mg\Delta h$$

$$P = \frac{\Delta E}{\Delta t}$$

$$\sum \frac{1}{2}mv_{\text{before}}^2 = \sum \frac{1}{2}mv_{\text{after}}^2$$

$$\Delta\vec{p} = \vec{F}_{\text{net}}\Delta t$$

$$\omega = \frac{\Delta\theta}{t}$$

$$\tau = r_{\perp}F = rF\sin\theta$$

$$v = \frac{2\pi r}{T}$$

$$U = -\frac{GMm}{r}$$

$$v = u + at$$

$$\vec{F}_{\text{net}} = m\vec{a}$$

$$W = F_{\parallel}s = Fs\cos\theta$$

$$K = \frac{1}{2}mv^2$$

$$P = F_{\parallel}v = Fv\cos\theta$$

$$\sum m\vec{v}_{\text{before}} = \sum m\vec{v}_{\text{after}}$$

$$a_c = \frac{v^2}{r}$$

$$F_c = \frac{mv^2}{r}$$

$$F = \frac{GMm}{r^2}$$

$$\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$$

### Waves and thermodynamics

$$v = f\lambda$$

$$f = \frac{1}{T}$$

$$d\sin\theta = m\lambda$$

$$n_x = \frac{c}{v_x}$$

$$I = I_{\max}\cos^2\theta$$

$$Q = mc\Delta T$$

$$f_{\text{beat}} = \left| f_2 - f_1 \right|$$

$$f' = f\frac{\left(v_{\text{wave}} + v_{\text{observer}}\right)}{\left(v_{\text{wave}} - v_{\text{source}}\right)}$$

$$n_1\sin\theta_1 = n_2\sin\theta_2$$

$$\sin\theta_c = \frac{n_2}{n_1}$$

$$I_1r_1^2 = I_2r_2^2$$

$$\frac{Q}{t} = \frac{kA\Delta T}{d}$$

FORMULAE SHEET (continued)

## Electricity and magnetism

$$E = \frac{V}{d}$$

$$V = \frac{\Delta U}{q}$$

$$W = qV$$

$$W = qEd$$

$$B = \frac{\mu_0 I}{2\pi r}$$

$$B = \frac{\mu_0 N I}{L}$$

$$\Phi = B_{||}A = BA\cos\theta$$

$$\varepsilon = -N\frac{\Delta\Phi}{\Delta t}$$

$$\frac{V_p}{V_s} = \frac{N_p}{N_s}$$

$$\vec{F} = q\vec{E}$$

$$F = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}$$

$$I = \frac{q}{t}$$

$$V = IR$$

$$P = VI$$

$$F = qv_{\perp}B = qvB\sin\theta$$

$$F = lI_{\perp}B = lIB\sin\theta$$

$$\frac{F}{l} = \frac{\mu_0}{2\pi}\frac{I_1 I_2}{r}$$

$$\tau = nIA_{\perp}B = nIAB\sin\theta$$

$$V_p I_p = V_s I_s$$

## Quantum, special relativity and nuclear

$$\lambda = \frac{h}{mv}$$

$$K_{max} = hf - \phi$$

$$\lambda_{max} = \frac{b}{T}$$

$$E = mc^2$$

$$E = hf$$

$$\frac{1}{\lambda} = R\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

$$t = \frac{t_0}{\sqrt{\left(1 - \frac{v^2}{c^2}\right)}}$$

$$l = l_0\sqrt{\left(1 - \frac{v^2}{c^2}\right)}$$

$$p_v = \frac{m_0 v}{\sqrt{\left(1 - \frac{v^2}{c^2}\right)}}$$

$$N_t = N_0 e^{-\lambda t}$$

$$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$$



## PERIODIC TABLE OF THE ELEMENTS

KEY

Atomic Number
Symbol
Standard Atomic Weight
Name

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1<br>H<br>1.008<br>Hydrogen | | | | | | | | | | | | | | | | | 2<br>He<br>4.003<br>Helium |
| 3<br>Li<br>6.941<br>Lithium | 4<br>Be<br>9.012<br>Beryllium | | | | | | | | | | | 5<br>B<br>10.81<br>Boron | 6<br>C<br>12.01<br>Carbon | 7<br>N<br>14.01<br>Nitrogen | 8<br>O<br>16.00<br>Oxygen | 9<br>F<br>19.00<br>Fluorine | 10<br>Ne<br>20.18<br>Neon |
| 11<br>Na<br>22.99<br>Sodium | 12<br>Mg<br>24.31<br>Magnesium | | | | | | | | | | | 13<br>Al<br>26.98<br>Aluminium | 14<br>Si<br>28.09<br>Silicon | 15<br>P<br>30.97<br>Phosphorus | 16<br>S<br>32.07<br>Sulfur | 17<br>Cl<br>35.45<br>Chlorine | 18<br>Ar<br>39.95<br>Argon |
| 19<br>K<br>39.10<br>Potassium | 20<br>Ca<br>40.08<br>Calcium | 21<br>Sc<br>44.96<br>Scandium | 22<br>Ti<br>47.87<br>Titanium | 23<br>V<br>50.94<br>Vanadium | 24<br>Cr<br>52.00<br>Chromium | 25<br>Mn<br>54.94<br>Manganese | 26<br>Fe<br>55.85<br>Iron | 27<br>Co<br>58.93<br>Cobalt | 28<br>Ni<br>58.69<br>Nickel | 29<br>Cu<br>63.55<br>Copper | 30<br>Zn<br>65.38<br>Zinc | 31<br>Ga<br>69.72<br>Gallium | 32<br>Ge<br>72.64<br>Germanium | 33<br>As<br>74.92<br>Arsenic | 34<br>Se<br>78.96<br>Selenium | 35<br>Br<br>79.90<br>Bromine | 36<br>Kr<br>83.80<br>Krypton |
| 37<br>Rb<br>85.47<br>Rubidium | 38<br>Sr<br>87.61<br>Strontium | 39<br>Y<br>88.91<br>Yttrium | 40<br>Zr<br>91.22<br>Zirconium | 41<br>Nb<br>92.91<br>Niobium | 42<br>Mo<br>95.96<br>Molybdenum | 43<br>Tc<br>Technetium | 44<br>Ru<br>101.1<br>Ruthenium | 45<br>Rh<br>102.9<br>Rhodium | 46<br>Pd<br>106.4<br>Palladium | 47<br>Ag<br>107.9<br>Silver | 48<br>Cd<br>112.4<br>Cadmium | 49<br>In<br>114.8<br>Indium | 50<br>Sn<br>118.7<br>Tin | 51<br>Sb<br>121.8<br>Antimony | 52<br>Te<br>127.6<br>Tellurium | 53<br>I<br>126.9<br>Iodine | 54<br>Xe<br>131.3<br>Xenon |
| 55<br>Cs<br>132.9<br>Caesium | 56<br>Ba<br>137.3<br>Barium | 57–71<br>Lanthanoids | 72<br>Hf<br>178.5<br>Hafnium | 73<br>Ta<br>180.9<br>Tantalum | 74<br>W<br>183.9<br>Tungsten | 75<br>Re<br>186.2<br>Rhenium | 76<br>Os<br>190.2<br>Osmium | 77<br>Ir<br>192.2<br>Iridium | 78<br>Pt<br>195.1<br>Platinum | 79<br>Au<br>197.0<br>Gold | 80<br>Hg<br>200.6<br>Mercury | 81<br>Tl<br>204.4<br>Thallium | 82<br>Pb<br>207.2<br>Lead | 83<br>Bi<br>209.0<br>Bismuth | 84<br>Po<br>Polonium | 85<br>At<br>Astatine | 86<br>Rn<br>Radon |
| 87<br>Fr<br>Francium | 88<br>Ra<br>Radium | 89–103<br>Actinoids | 104<br>Rf<br>Rutherfordium | 105<br>Db<br>Dubnium | 106<br>Sg<br>Seaborgium | 107<br>Bh<br>Bohrium | 108<br>Hs<br>Hassium | 109<br>Mt<br>Meitnerium | 110<br>Ds<br>Darmstadtium | 111<br>Rg<br>Roentgenium | 112<br>Cn<br>Copernicium | 113<br>Nh<br>Nihonium | 114<br>Fl<br>Flerovium | 115<br>Mc<br>Moscovium | 116<br>Lv<br>Livermorium | 117<br>Ts<br>Tennessine | 118<br>Og<br>Oganesson |

Lanthanoids

| | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 57<br>La<br>138.9<br>Lanthanum | 58<br>Ce<br>140.1<br>Cerium | 59<br>Pr<br>140.9<br>Praseodymium | 60<br>Nd<br>144.2<br>Neodymium | 61<br>Pm<br>Promethium | 62<br>Sm<br>150.4<br>Samarium | 63<br>Eu<br>152.0<br>Europium | 64<br>Gd<br>157.3<br>Gadolinium | 65<br>Tb<br>158.9<br>Terbium | 66<br>Dy<br>162.5<br>Dysprosium | 67<br>Ho<br>164.9<br>Holmium | 68<br>Er<br>167.3<br>Erbium | 69<br>Tm<br>168.9<br>Thulium | 70<br>Yb<br>173.1<br>Ytterbium | 71<br>Lu<br>175.0<br>Lutetium |

Actinoids

| | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 89<br>Ac<br>Actinium | 90<br>Th<br>232.0<br>Thorium | 91<br>Pa<br>231.0<br>Protactinium | 92<br>U<br>238.0<br>Uranium | 93<br>Np<br>Neptunium | 94<br>Pu<br>Plutonium | 95<br>Am<br>Americium | 96<br>Cm<br>Curium | 97<br>Bk<br>Berkelium | 98<br>Cf<br>Californium | 99<br>Es<br>Einsteinium | 100<br>Fm<br>Fermium | 101<br>Md<br>Mendelevium | 102<br>No<br>Nobelium | 103<br>Lr<br>Lawrencium |

Standard atomic weights are abridged to four significant figures.
Elements with no reported values in the table have no stable nuclides.
Information on elements with atomic numbers 113 and above is sourced from the International Union of Pure and Applied Chemistry Periodic Table of the Elements (November 2016 version).
The International Union of Pure and Applied Chemistry Periodic Table of the Elements (February 2010 version) is the principal source of all other data. Some data may have been modified.